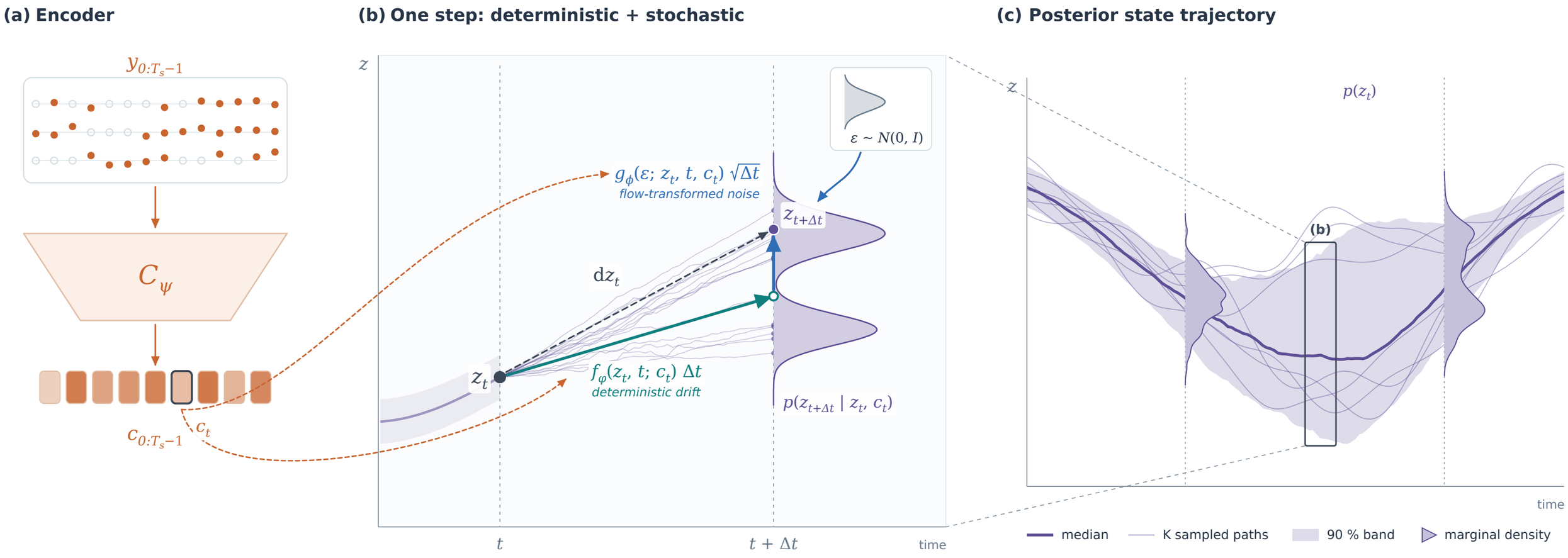

(a) Encoder
$y_{0:T_s-1}$
C_ψ
$c_{0:T_s-1}$
c_t
(b) One step: deterministic + stochastic
z
$\varepsilon \sim N(0, I)$
$g_\phi(\varepsilon;\, z_t,\, t,\, c_t)\,\sqrt{\Delta t}$
flow-transformed noise
$z_{t+\Delta t}$
$\mathrm{d}z_t$
z_t
$f_\varphi(z_t,\, t;\, c_t)\,\Delta t$
deterministic drift
$p(z_{t+\Delta t} \mid z_t,\, c_t)$
t
$t + \Delta t$
time
(c) Posterior state trajectory
z
$p(z_t)$
(b)
time
median
K sampled paths
90 % band
marginal density